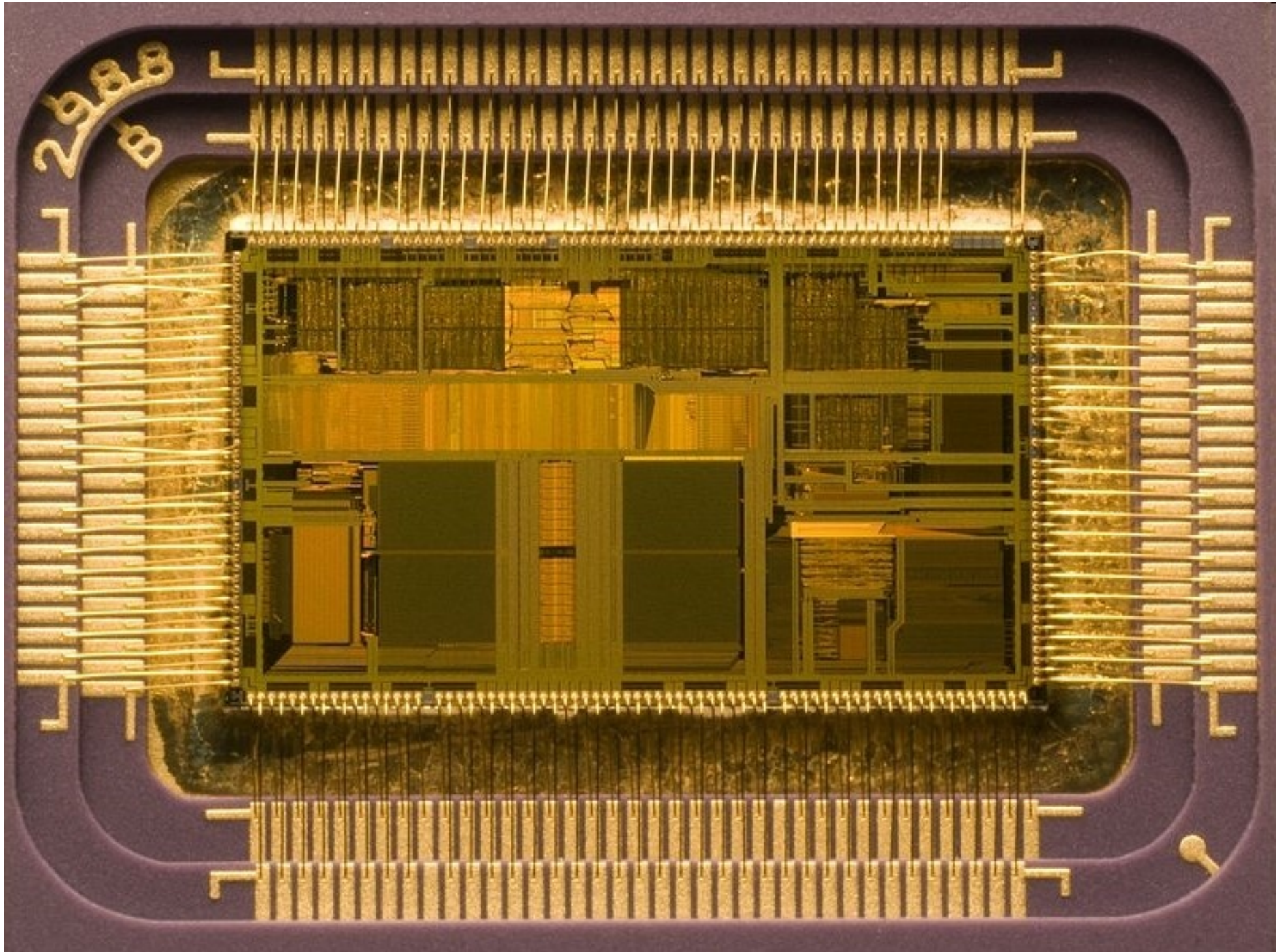


Advanced Micro-Controller Systems



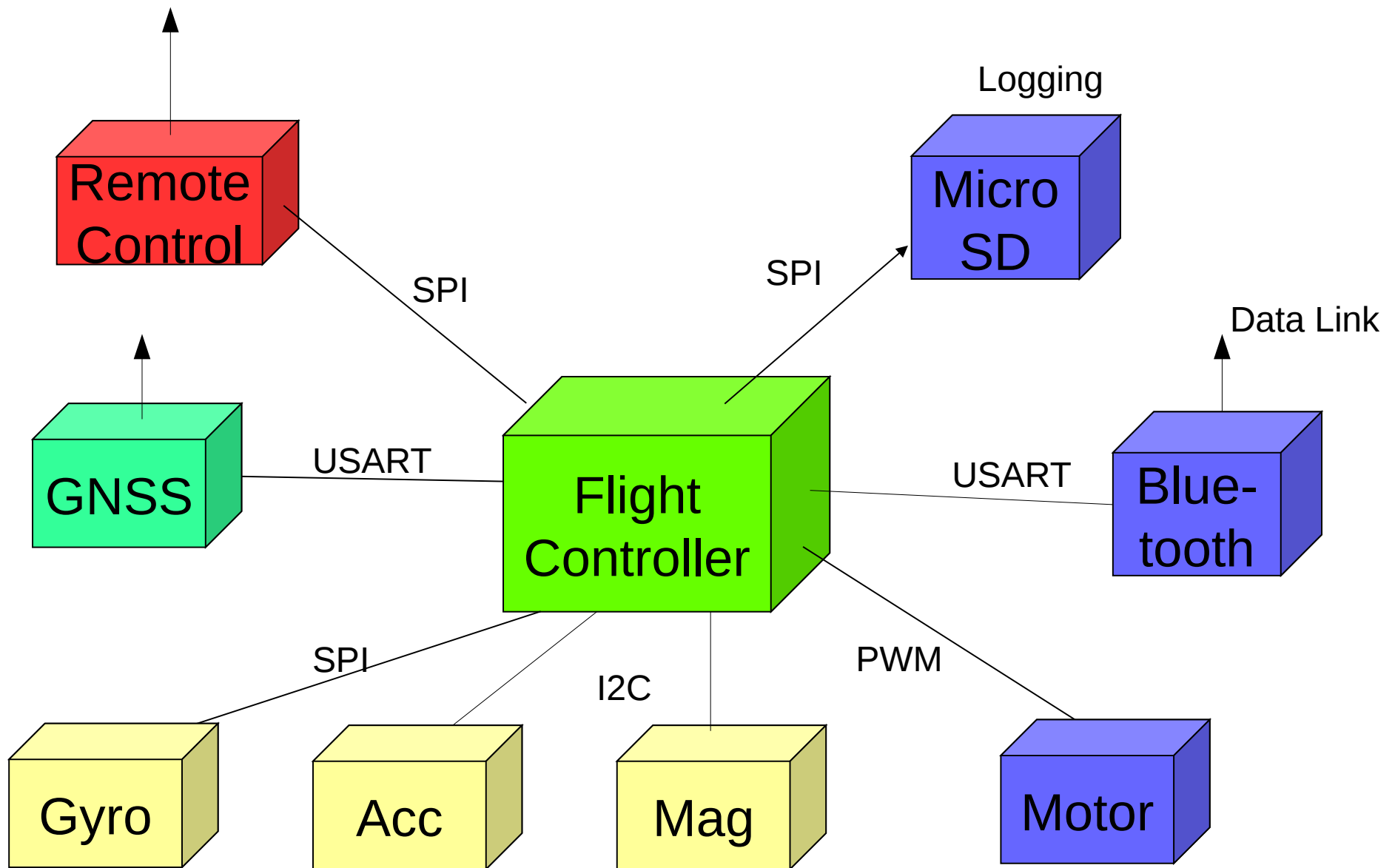
INTEL 80286 PC-Processor



Project "Colibri"



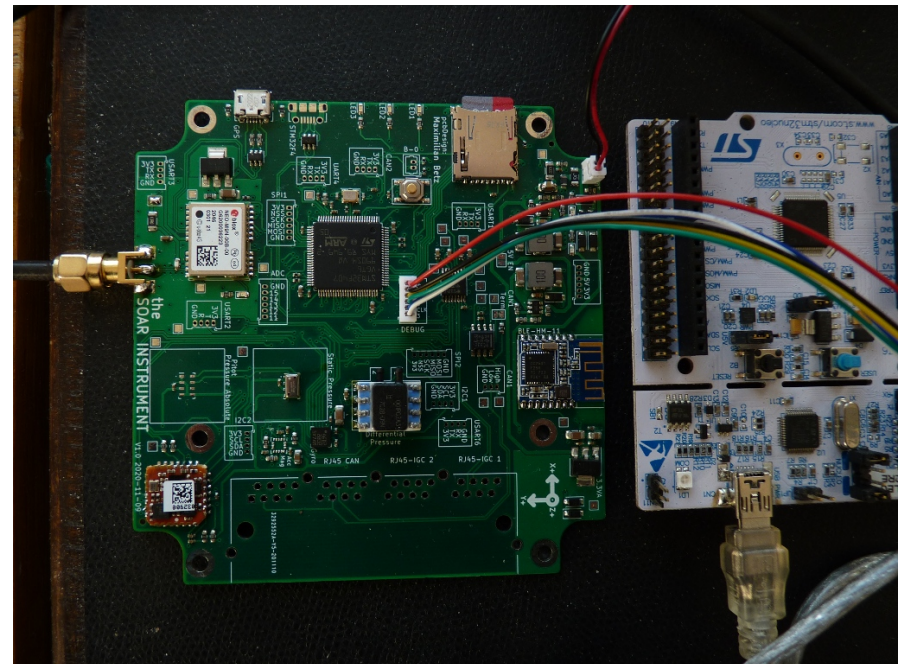
„Colibri“ Control System



Octocopter "Colibri Epsilon"



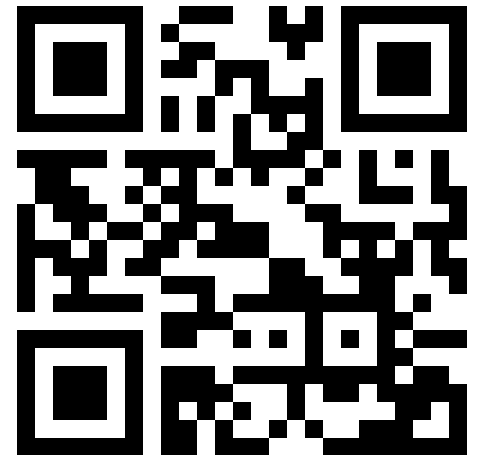
Project "Albatros MK2"



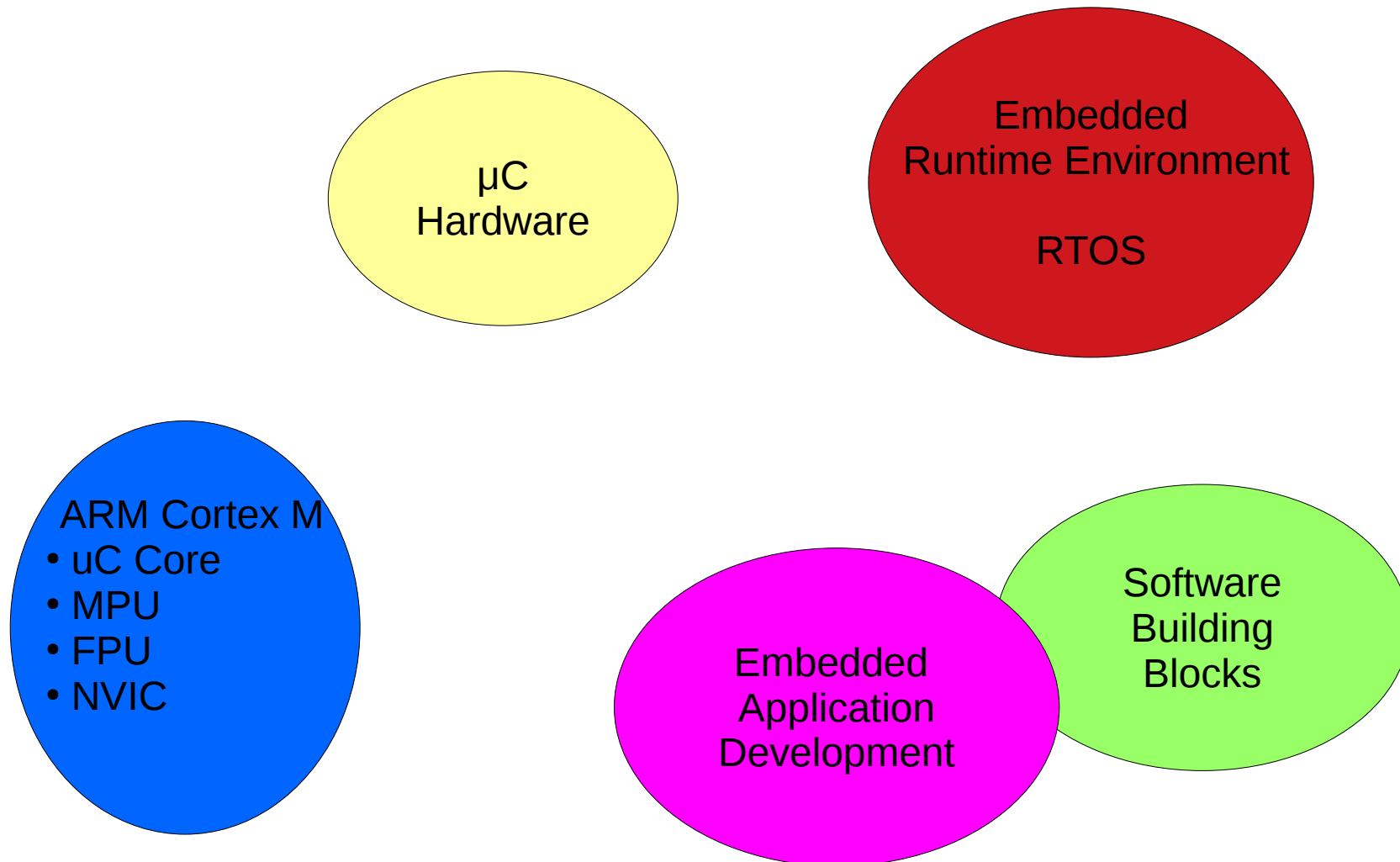
AMS Module definition

Content of the course Advanced Micro-Controller Systems

- Hardware Architecture of current Micro-Controller Systems
- RTOS implementation
- Tasks, Events, Messages, Semaphores
- Critical Sections , Priority Ceiling, Deadlocks
- Scheduling algorithms
- Safety and Memory-Protection
- Hardware Security Features
- Hardware Device-Driver development
- Efficient Implementation of DSP algorithms



AMS Material

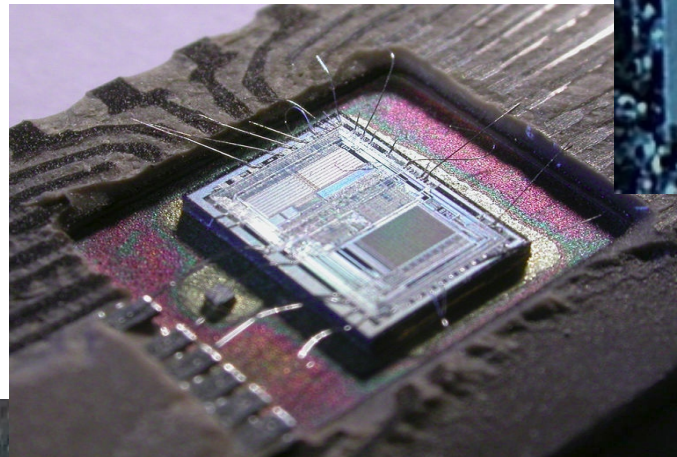


Advanced Micro-controller Systems

4004

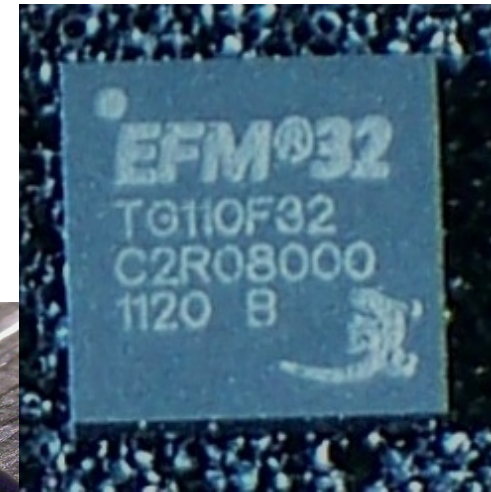


1971



8742

1976



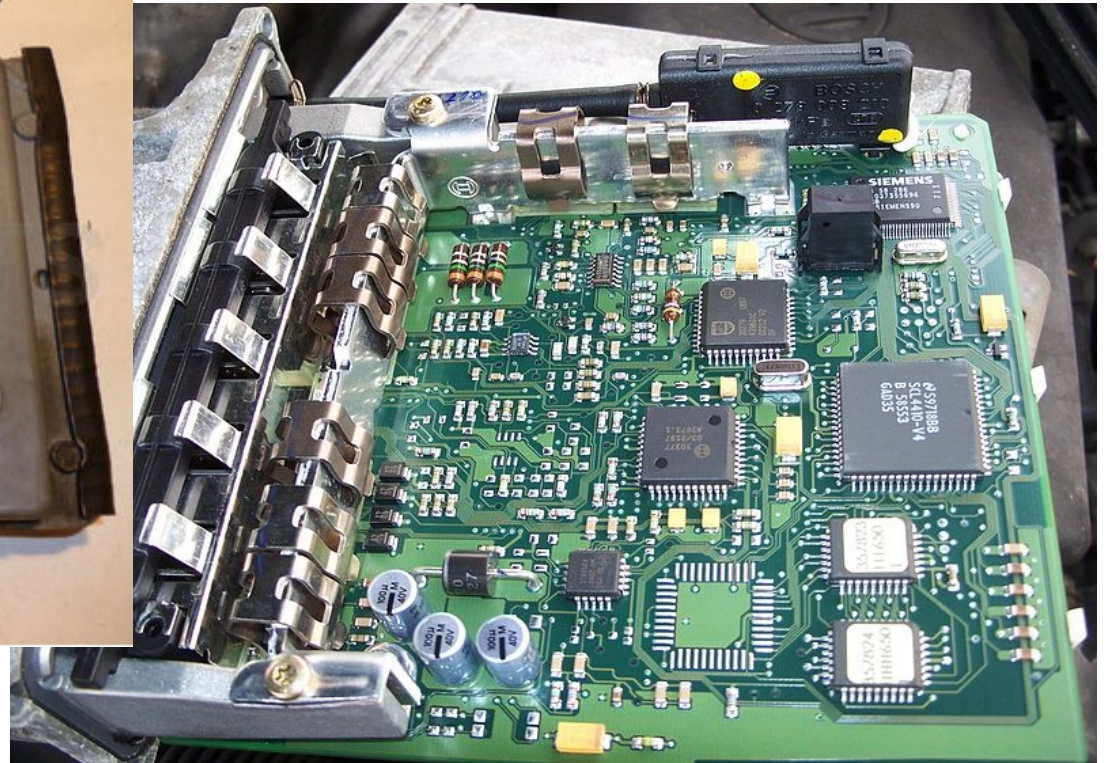
32-bit Controller

2005

Industrial Control Systems



Automotive Electronic Control Modules



Cost sensitive high volume applications
Digital signal processing DSP
Integrated power electronics

Avionics



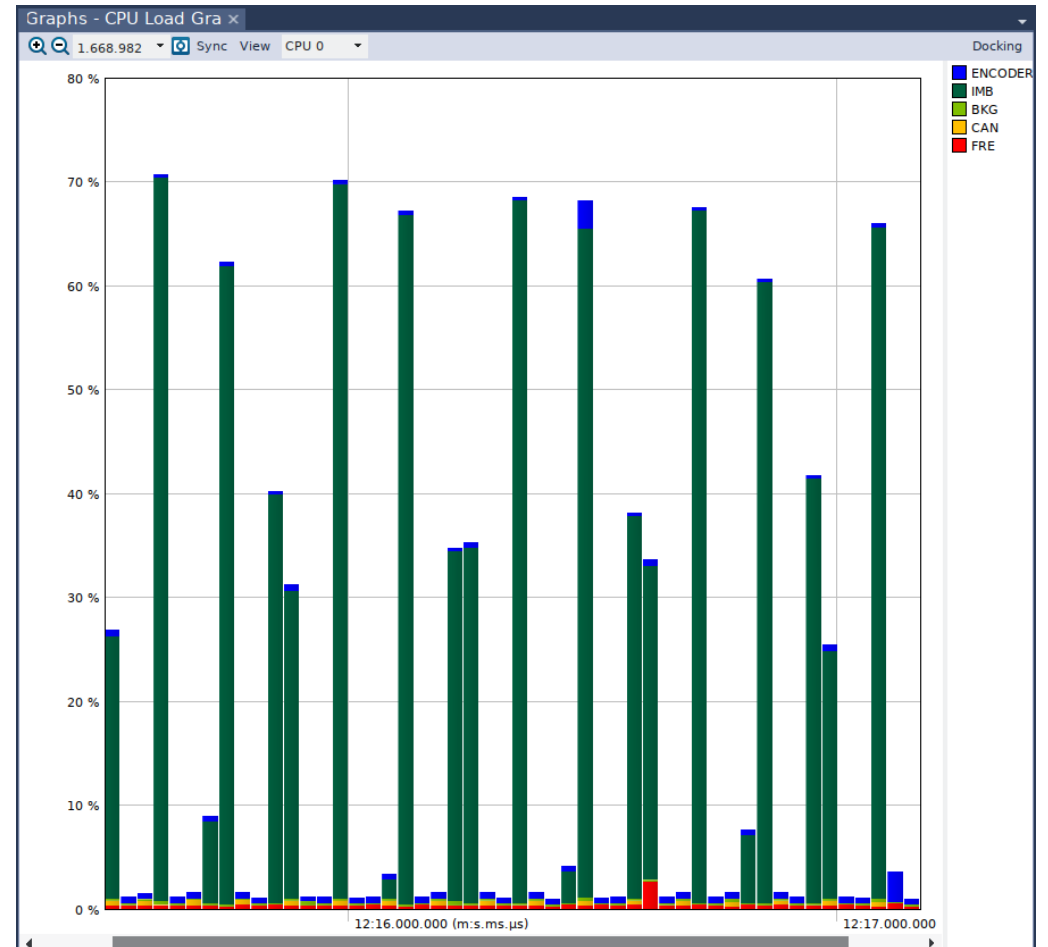
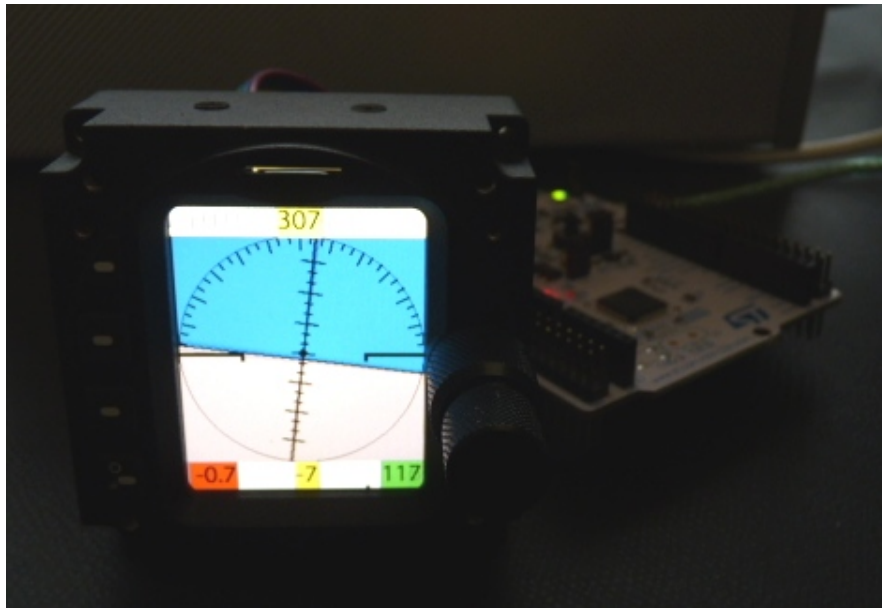
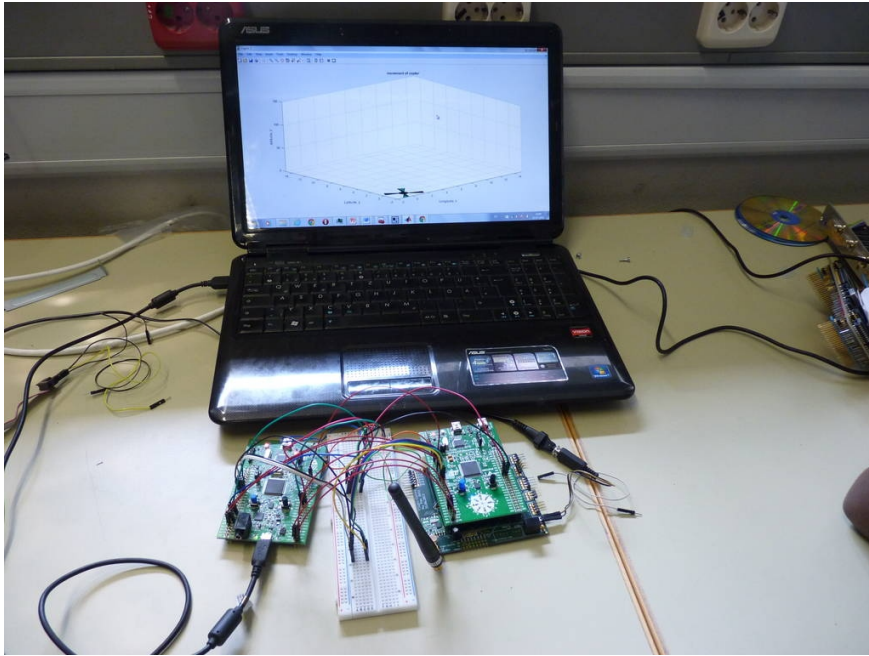
Demanding safety and security
Low lot size
Critical environment (temperature, vibration, radiation, etc.)

Smart Meters, IOT



Cost sensitive application
Very low power consumption
Officially calibratable
Remote sensing via UHF radio link

Application Development



AMS Lab: ???

Several problems are still unsolved:

- Can we organize a conventional lab at H_DA this semester ?

or

- Can we find &| buy enough uC boards for “lab at home” ?

or

- Shall we use the pSOC board for the AMS lab ?

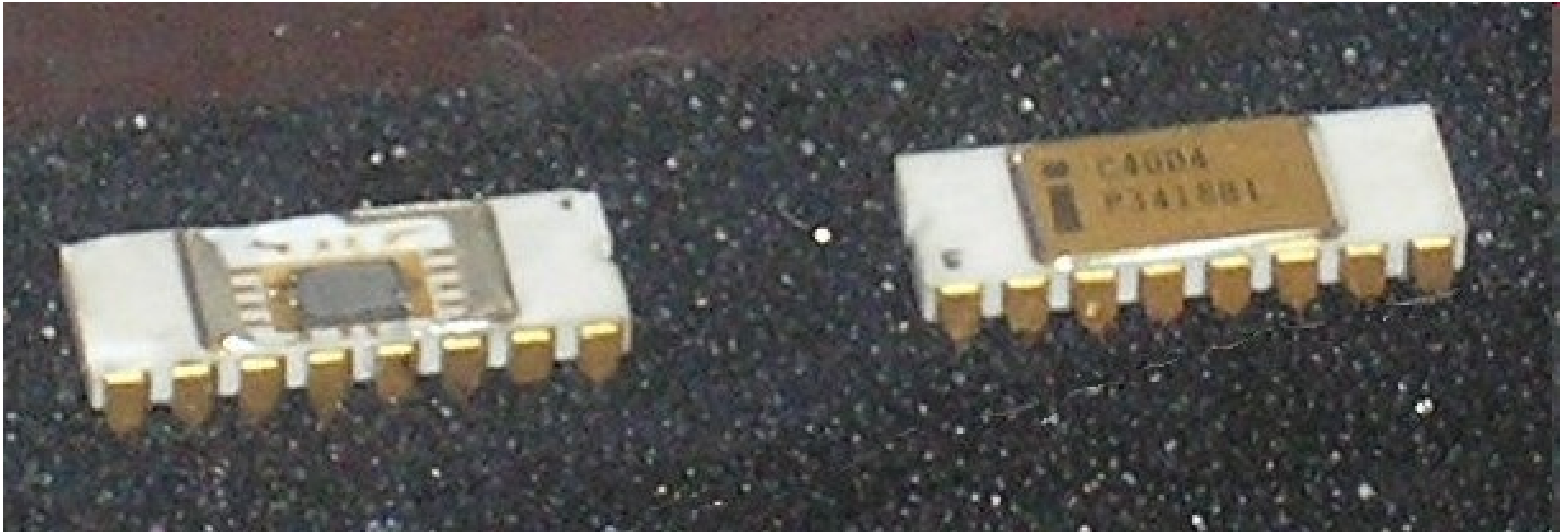
A solution will be published ASAP

Computer Architecture Milestones

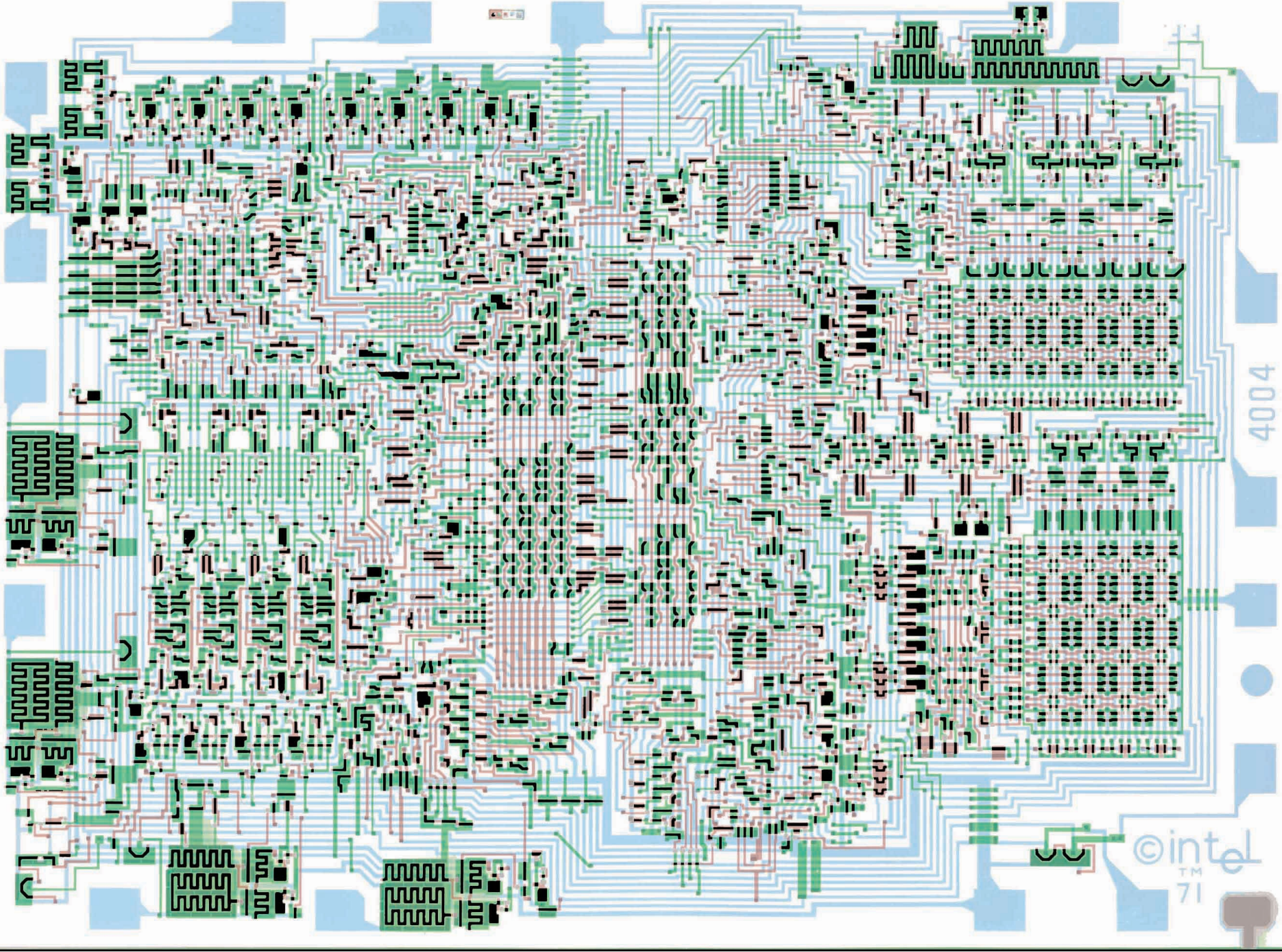
1834	Babbage	First try to build a digital computer
1936	C. Zuse	First scientific relay Computer
1943	Colossus	First electronic computer
1946	Eniac	First useful numerical computer
1951	Whirlwind	First real-time computer
1960	PDP1	First minicomputer (50 pcs sold)
1965	PDP8	Minicomputer (50.000 pcs sold)
1971	4004	First integrated microprocessor CPU
1981	IBM PC	Universal personal computer
1993	Newton	Palmtop computer



First Microprocessor



First 4 Bit Mikroprocessor: Intel 4004 (1971)



1976: First Microcontroller

INTEL 8742

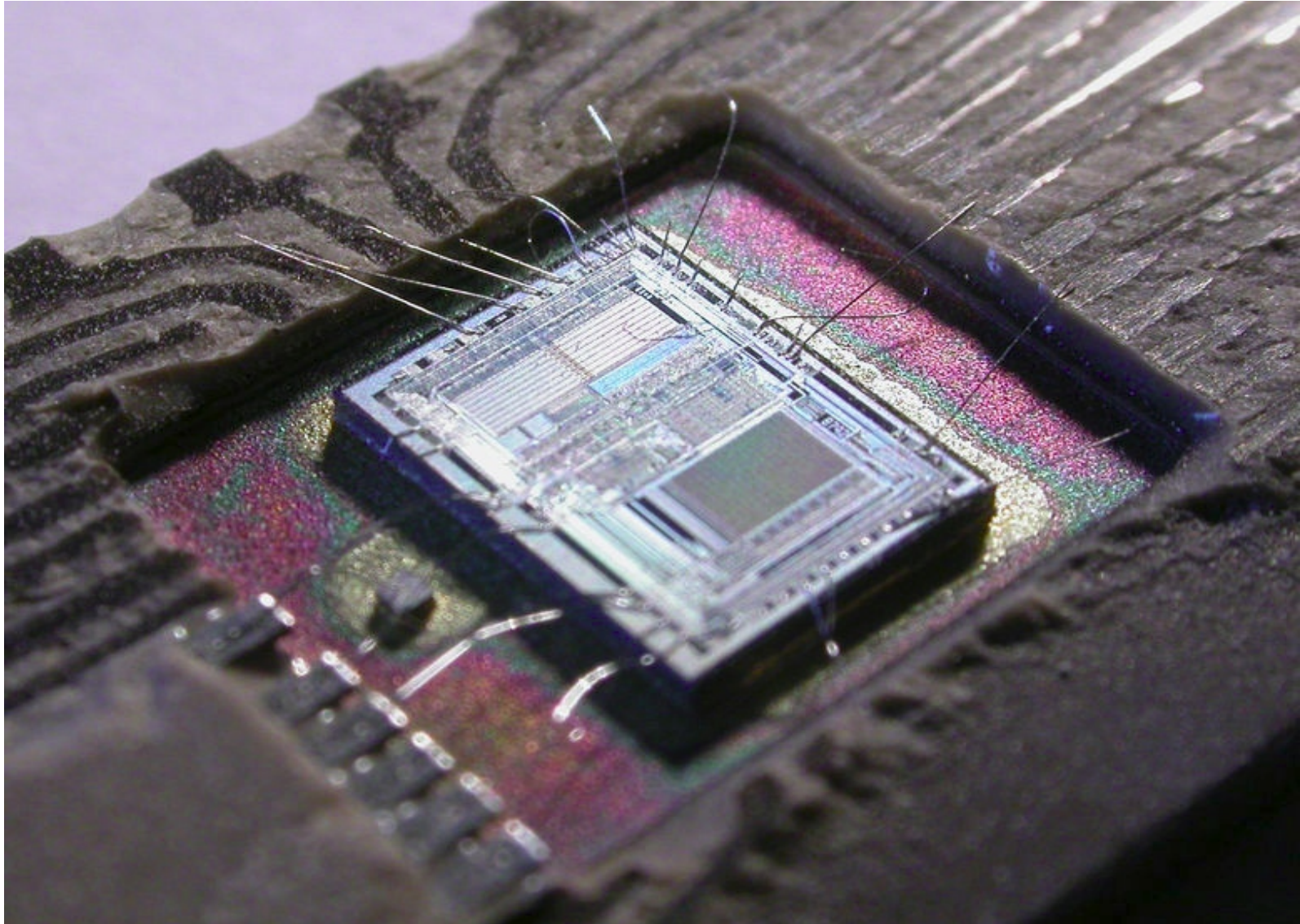
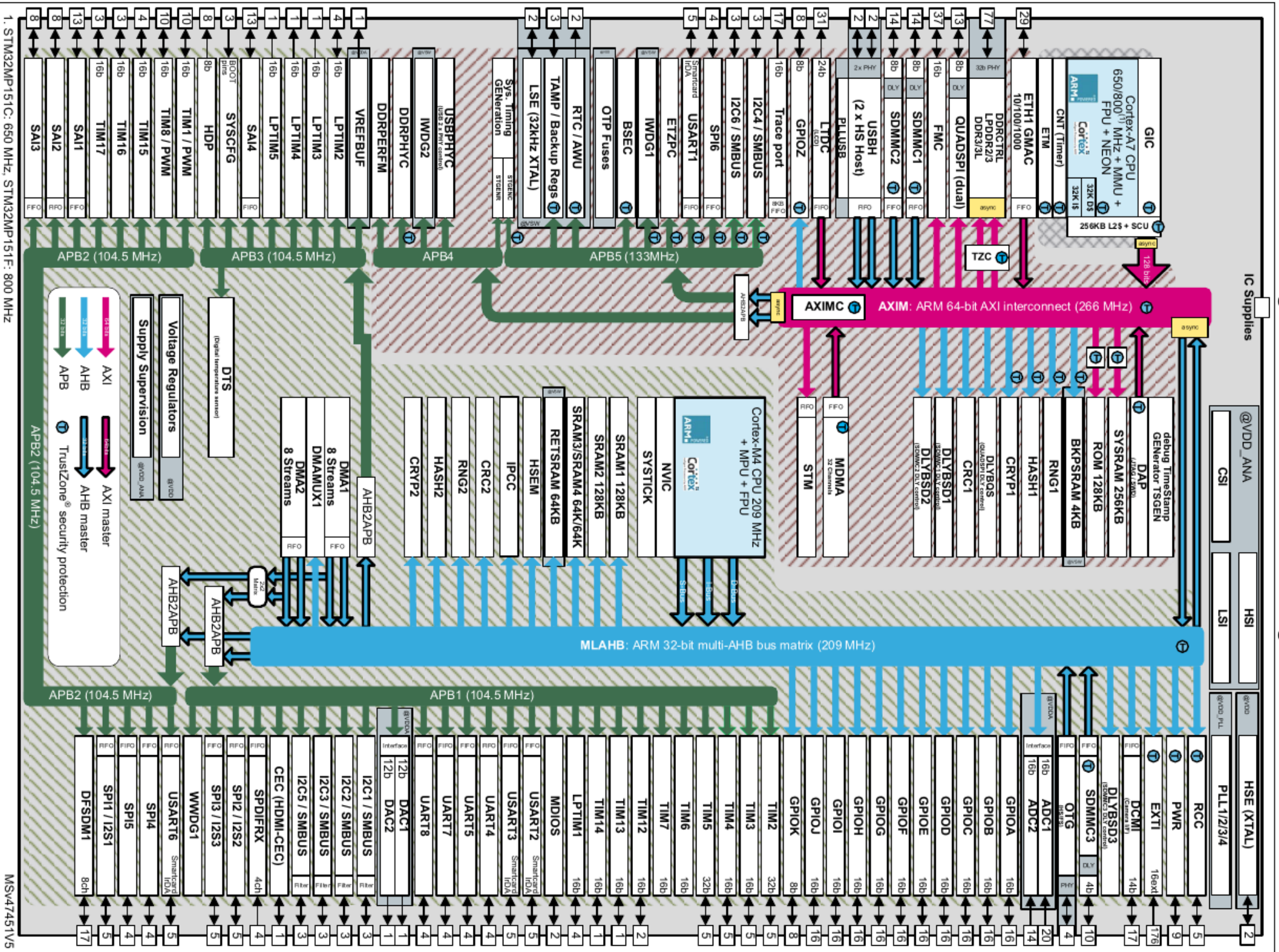
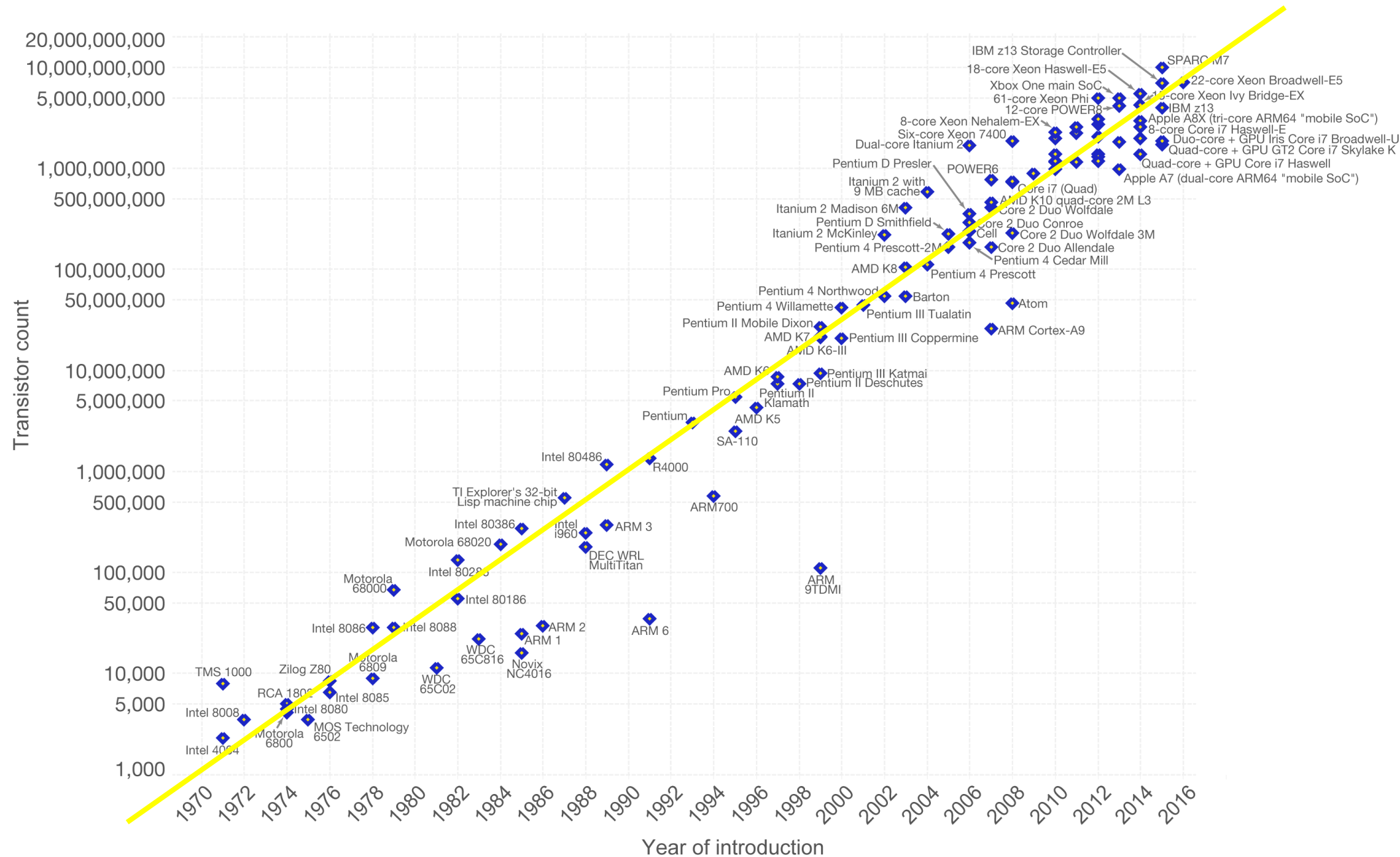


Figure 1. STM32MP151C/F block diagram



Moore's Law – The number of transistors on integrated circuit chips (1971-2016)

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are strongly linked to Moore's law.

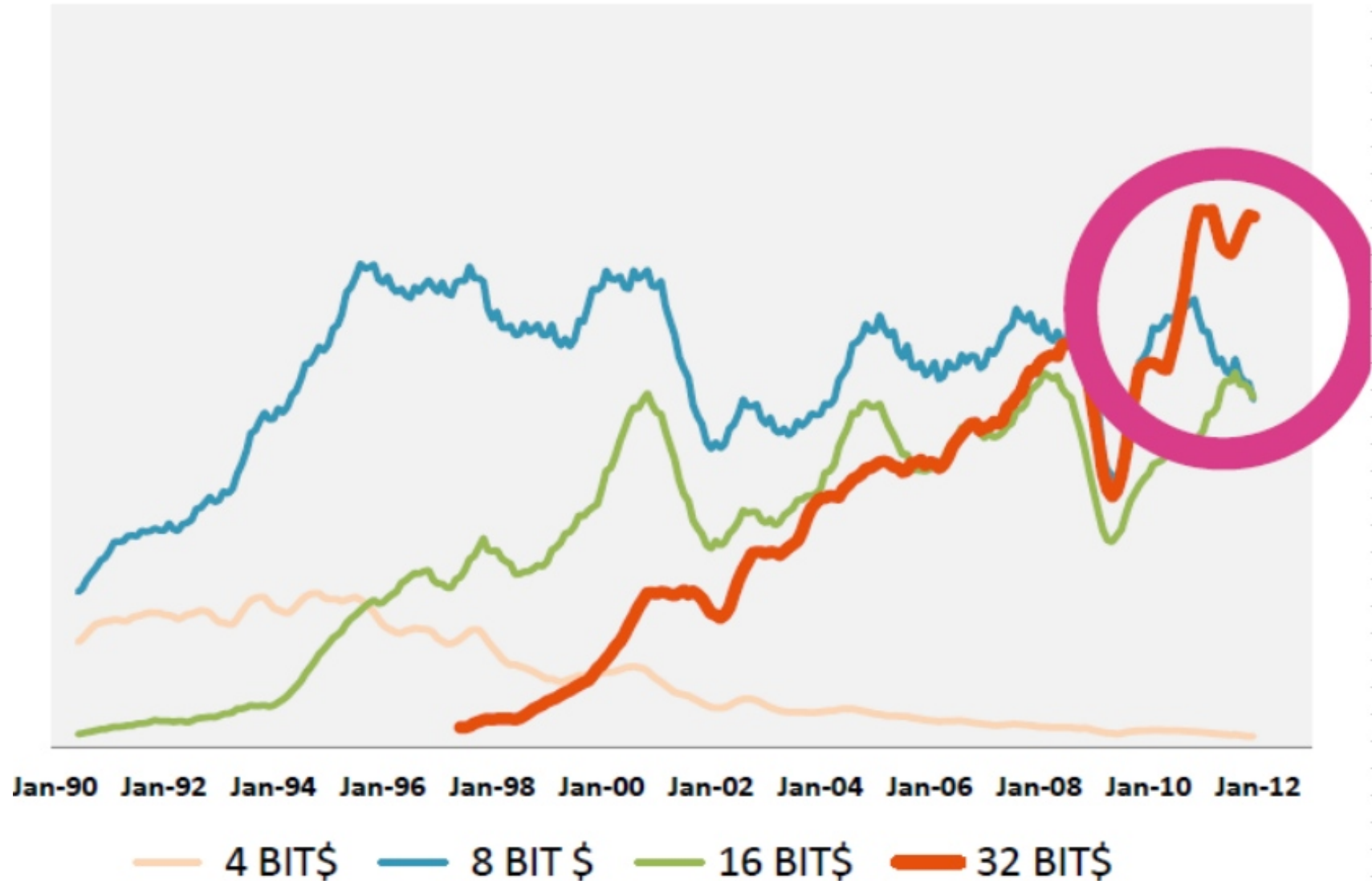


Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)

The data visualization is available at [OurWorldinData.org](https://www.ourworldindata.org). There you find more visualizations and research on this topic.

Licensed under [CC-BY-SA](https://creativecommons.org/licenses/by-sa/4.0/) by the author Max Roser.

CPU Architecture change



Source: WSTS (Worldwide Semiconductor Trade Statistics)

ARM Cortex M Manufacturers

Actel

INTEL

Silicon labs

Fujitsu

ATMEL/Microchip

Toshiba

Freescall

Cypress Semiconductor

Energy Micro

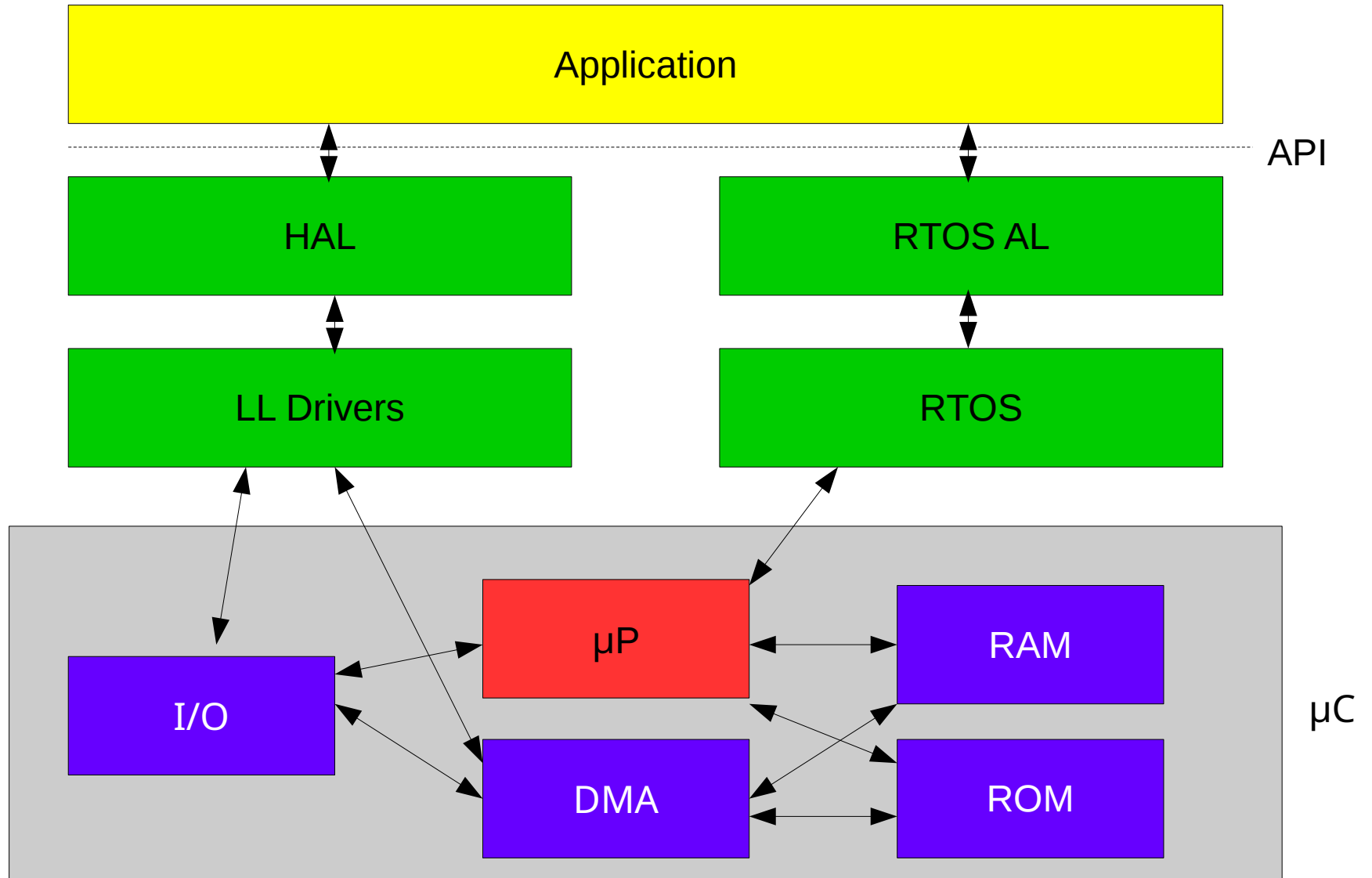
NXP

ST Microelectronics

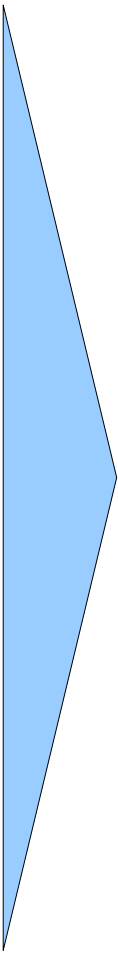
Texas Instruments

Next day probably
some more :-)

μC System



Lecture Overview

- 
- Introduction, Microcontroller basics
 - 32-bit Microcontroller technology
 - ARM Cortex M Architecture, (CM0 CM3 CM4 CM7)
 - ARM Cortex M Microcontroller Instruction Set
 - Exception Handling, Interrupts
 - Real-time Operating Systems (RTOS)
 - Memory Management, protected memory, MPU
 - Implementing Calculations, Data Formats, FPU
 - I/O, Device drivers
 - Application Design
 - Case Studies, Advanced Design Patterns

<https://skript.eit.h-da.de/ams/>

